

Assignment 1: Road Segmentation Using Naïve Bayes

- ❖ In this task you are required to apply naive bayes segmentation using the attached dataset. This dataset has five classes:
 1. **road_surface**: surface of the road → blue color in the mask
 2. **marking**: white and yellow marking on the road, → yellow color in the mask
 3. **road_sign**: road signs, → red color in the mask
 4. **car**: cars on the road, → orange color in the mask
 5. **background**: side of the road and surrounding objects → purple color in the mask
- ❖ In order to solve this, you will need to apply the following:
 1. Convert the data from the training images and masks to be in a format usable for naive bayes; to be used similar to the lab code.
 2. Use histogram-based naive Bayes or Gaussian-based naive bayes to extract statistics from the training data for five classes (You can use lab code as starting point and modify it, built-in naïve bayes is not allowed).
 3. Test the segmentation results using the image: “test.jpg”, the required output should be all the road pixels in white while everything else is black.
- ❖ You will submit a single .py file that has all your code and has the name StudentID.py
- ❖ **The assignment is individual, and cheating detection will be applied.**